

Research Plan

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Define your research question:

1. Exploring PPO activity in green and red apple extract using HPLC-based quantification of dopamine

What was your motivation for choosing this type of question?

1. Exploring the different amounts of PPO activity in different fruits

What type of experiments will you perform on each day of your project?

Day 1: Preparation of dilution series for dopamine

Use 10 mL volumetric flasks to prepare the following dilution series:

	Dopamine concentration (mg/mL)	Preparation
A	1.53	Already prepared (identical to stock solution)
B	0.765	Place 5 mL of the 1.53 mg/mL stock solution (A) into a 10 mL volumetric flask and fill up to the 10 mL mark with HPLC-grade water.
C	0.153	Place 1 mL of the 1.53 mg/mL stock solution (A) into a 10 mL volumetric flask and fill up to the 10 mL mark with HPLC-grade water.
D	0.077	Place 1 mL of the 0.765 mg/mL solution (from step B) into a 10 mL volumetric flask and fill up to the 10 mL mark with HPLC-grade water.
E	0.015	Place 1 mL of the 0.153 mg/mL solution (from step C) into a 10 mL volumetric flask and fill up to the 10 mL mark with HPLC-grade water.

Place the diluted dopamine solutions into labeled microcentrifuge tubes and spin the tubes at 15,000 rpm for 15 minutes. Transfer the clear supernatant into labeled HPLC vials.

B. Preparation of apple(green/red) crude extract w/o peel

- a. Weigh out 80-100g of apples and cut them into small pieces. Record the exact weight of apple pulp in your lab notebook. Mash these pieces at room temperature into a soggy paste. Homogenize the ground apple pulp in a small volume of methanol (apx. 10 mL, use as little as possible). Strain or drain off the liquid into a smaller beaker. Then transfer the liquid into microcentrifuge tubes and spin the tubes at 15,00 rpm for 15 minutes or

more. If the supernatant is clear , transfer it into a labeled HPLC vial. If not, transfer the supernatant into a fresh microcentrifuge tube and spin again.

Samples	Initial Weight (g)	Boiled	Water (mL)	Methanol (mL)	
Red	8.3572	No	-	5.0	
Red	10.6304	Yes	70	5.0	
Green	8.2199	No	-	5.0	
Green	10.5936	Yes	70	5.0	
Red Water	-	Yes	-	-	
Green Water	-	Yes	-	-	

C. HPLC

You will put together a method with the following parameters:

- Column: Zorbax Eclipse XDB C18, particle size: 5 μ m, dimensions: 150 x 4.6 mm
- Column temperature: 30°C (or make your own choice)
- Solvent A: 0.1 % formic acid in HPLC-grade water
- Solvent B: methanol
- Detector settings for absorbance: Wavelength 274 nm, bandwidth 4 nm (reference wavelength 360 nm with 60 nm bandwidth)
- Detector settings for fluorescence: excitation wavelength 279 nm, emission wavelength: 320 nm
- Injection volume 1 μ L (with needle wash and an extra methanol vial in position 91)
- Pump: Flow rate: 1 mL/min
- Gradient:

Injection volume 5 μ L (with needle wash and an extra methanol vial in position 91)

Timing	Solvent composition	Comment
Start (0 min)	Start with only solvent A (acidified water) or a low percentage of B (methanol).	The starting solvent should be polar, so that many molecules can stick to the column.

Can vary between 5-15 minutes depending on the complexity of the mixture	Your target molecule (dopamine) elutes at less than 10% methanol. You can experiment with a step-wise or smooth increase in methanol.	Your target molecule should elute in this part of the program.
At least 5 minutes	100 % B	Since apple pulp extract contains many molecules it is important to rinse off any hydrophobic (sticky) molecules with a high percentage of B before starting a new run
10 minutes or longer	Your starting condition (for example only solvent A)	Re-equilibrate the column into starting conditions

The HPLC instrument will be programmed to run a sequence with your methods for the standards and samples. The data will be available next week for analysis.

Day 2: Data Analysis

- A. Inspect the chromatograms of your dopamine standard samples. The largest peak in these chromatograms should belong to dopamine. As the dopamine concentration decreases, the peak area and peak height of the main peak should decrease as well. Make a table in which you record the retention time and peak area for each standard solution.

Sample	Chlorogenic acid Concentration (mg/mL)	Retention Time (min) at 320	Retention Time (min) at 360	Peak Area (LU x Sec)
	Apple trace at specific wavelength	Peak saturation 320		
A	1.53			
B	.765			
C	.153			
F	.077			

	.015 E			
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Check for consistent retention times for the dopamine peak. Calculate the average and standard deviation or identify the range of values, so that you can report on the consistency between individual HPLC runs.

Next, create a calibration curve by plotting the values for the peak area on the y-axis and the concentration on the x-axis. The trendline equation should go through zero as there should be no peak in the absence of dopamin. The slope of the trendline equation will be used to calculate the concentration of dopamine in your banana extract samples. The R-square value of your trendline should be close to 1 (values ranging from 0.995 to 0.998 are typical). If your R-square value is smaller than 0.95, you probably were not careful enough in preparing your dilution series. Please check for mistakes in extracting data from the HPLC chromatograms.

To identify dopamine in the HPLC chromatograms of the apple extract, create an overlay using a chromatogram of a banana extract and a domain standard. The chromatogram of the banana extract will contain numerous peaks. Their peak corresponding to dopamine should have the same retention time as in the standards. Create a table in your lab notebook with the following enteries:

Trail 1:

Sample	Extraction ratio (g apple peel per mL methanol)	Retention time of Chlorogenic acid peak at 320 (abs/min)	Retention time of Chlorogenic acid peak at 360 (abs/min)	Boiled	Peak area of chlorogenic acid peak (LU x sec) at 320 (abs/min)	Peak area of chlorogenic acid peak (LU x sec) at 360 (abs/min)	Calculating concentration of Chlorogenic acid in units of mg/mL	Chlorogenic acid amount in mg per 100g apple pulp
Red		15.5	15.5	No	7533.1	1926.7		
Red		15.505	15.505	Yes	1061.1	266.2		
Green		15.508	15.508	No	516.1	130.5		
Green		15.508	N/A	Yes	181.3	N/A		
Red Water		15.505	15.505	Yes	1061.1	266.2		
Green Water		15.508	N/A	Yes	181.3	N/A		

Trial Number 2:

Sample	Extraction ratio (g apple peel per mL methanol)	Retention time of Chlorogenic acid peak at 320 (abs/min)	Retention time of Chlorogenic acid peak at 360 (abs/min)	Boiled	Peak area of dopamine peak (LU x sec) at 320 (abs/min)	Peak area of dopamine peak (LU x sec) at 360 (abs/min)	Calculating concentration of dopamine in units of mg/mL	Dopamine amount in mg per 100g apple pulp
Red		15.491	15.491	No	7494.8	1936.5		
Red		15.495	15.495	Yes	1043.3	264.2		
Green		15.501	15.501	No	508.4	121.6		
Green		15.494	15.494	Yes	180.3	42.8		
Red Water		15.48	15.479	Yes	1258	316.5		
Green Water		15.493	15.493	Yes	222.6	52.8		

The concentration of dopamine in your apple extract samples can be calculated with the following formula:

$$\text{Concentration of dopamine } \left(\frac{\text{mg}}{\text{mL}}\right) = \frac{\text{Peak area of dopamine peak in chromatogram of banana extract}}{\text{Slope of calibration curve (see trendline for dopamine standards)}}$$

To determine the Dopamine amount per 100g apple pulp you need to divide the dopamine concentration (mg/mL) by the extraction ratio (g banana/mL extraction solvent) and multiply by 100.

$$\text{Dopamine amount (in mg) per 100 g banana peel or pulp} = \frac{100 \times \text{Concentration of dopamine } \left(\frac{\text{mg}}{\text{mL}}\right)}{\text{Extraction ratio } \left(\frac{\text{g banana peel or pulp}}{\text{mL methanol}}\right)}$$

Day 3: To be determined as the experiment progresses.

List of supplies:

- 10 mL volumetric flasks (x4)
- 100 mL volumetric flask
- Microcentrifuge tubes
- mortar/pestle
- Beaker (50 mL)
- Green apple
- Red apple

List of chemicals:

- Methanol (20 mL)
- HPLC-grade water (350 ml)
- Dopamine hydrochloride (0.190 g)

Hazards:

- Dopamine hydrochloride - Irritant and environmental hazard
- Methanol - Flammable, acute toxic, and health hazard